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Oefeningenreeks 5B nr. 1.12.

$$\int_{-\infty}^{0} xe^x dx$$

$$= \lim_{a \to -\infty} \int_a^0 xe^x$$

$$\begin{cases} u = x \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = dx \\ v = e^x \end{cases}$$

$$= \lim_{a \to -\infty} \left(xe^x - \int_a^0 e^x dx \right)$$

$$= \lim_{a \to -\infty} [xe^x - e^x]_a^0$$

$$= \lim_{a \to -\infty} (0e^0 - e^0) - (ae^a - e^a)$$

$$= -1 + \infty \cdot e^{-\infty} + e^{-\infty}$$

$$= -1$$

References